7 ลูก ขาว 3 ดำ 4

(a) สีต่างกัน

$$\Omega = 7 \times 6 = 42$$

$$W = \binom{3}{1}\binom{2}{1} = 6$$

$$b = \binom{4}{1}\binom{3}{1} = 12$$

$$\therefore \frac{42-(6+12)}{42} = \frac{\cancel{24}\ \cancel{12}\ 4}{\cancel{42}\ \cancel{21}\ 7} = \frac{4}{7} \ \#$$

(b) สีเดียวกัน

$$\Omega = 7 \times 6 \times 5 = 210$$

$$W = \binom{3}{1}\binom{2}{1}\binom{1}{1} = 6$$

$$b = \binom{4}{1}\binom{3}{1}\binom{2}{1} = 24$$

$$\therefore \frac{6+24}{210} = \frac{\cancel{30}\ \cancel{10}\ 1}{\cancel{210}\ \cancel{70}\ 7} = \frac{1}{7} \ \#$$

(c) เหรียญออกหน้า

$$\Omega = \binom{7}{3} = \frac{5040}{144} = 35$$

$$A = \binom{3}{3} + \binom{4}{3} = 1 + \frac{24}{6} = 1 + 4 = 5$$

$$\therefore \frac{\cancel{5}\ 1}{\cancel{35}\ 7} = \frac{1}{7} \ \#$$